

SCHOOL LEARNING PLATFORM

Project Planning Document

Phase 1: Research — Month 1 (User Flow, User Stories & API Design)

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JavaScript Project — TPC Performance Centre

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1. Project Overview

This document consolidates the Month 1 research and planning work for the school learning platform: the initial project brief, the user-flow diagrams, the full set of user stories for both user roles, the REST API design, and the earliest hand-drawn wireframes.

The application is being designed for high school users, with two distinct roles that share one codebase but see different navigation and screens: the Learner and the Assessor/Lecturer.

Planning approach

User stories follow the standard syntax:

As a [role], I want/must [goal], so that [benefit].

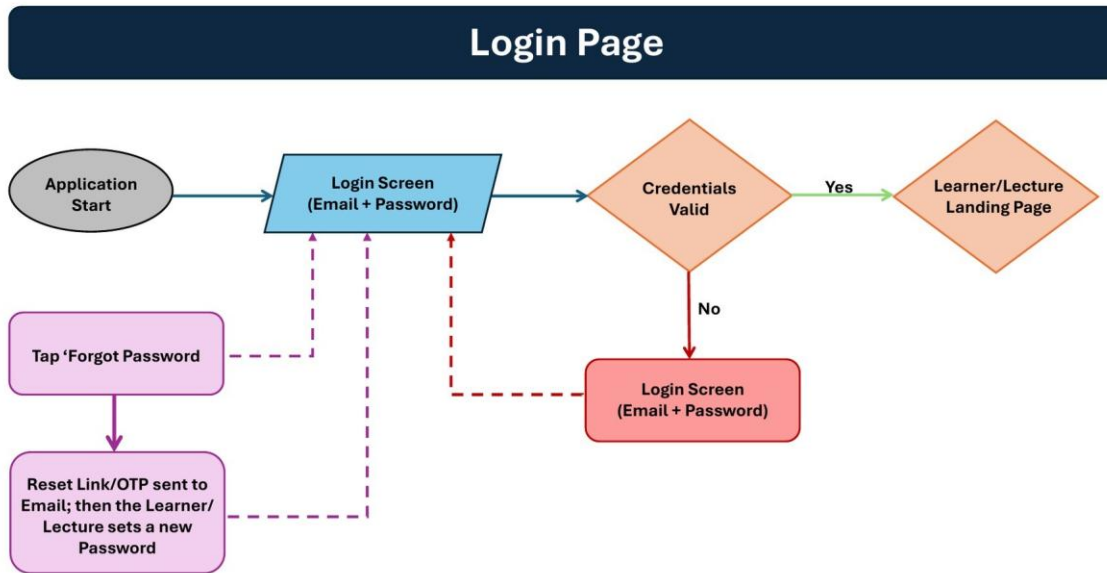
The Month 1 goal was to plan and design the Learner Dashboard first, covering the following areas:

- Authentication & Profile
- Assignments & Assessments
- Notifications & Reminders
- Achievements & Motivation
- Course Progress
- Navigation & Usability
- Assessor / Lecturer role

2. User Flow Diagrams

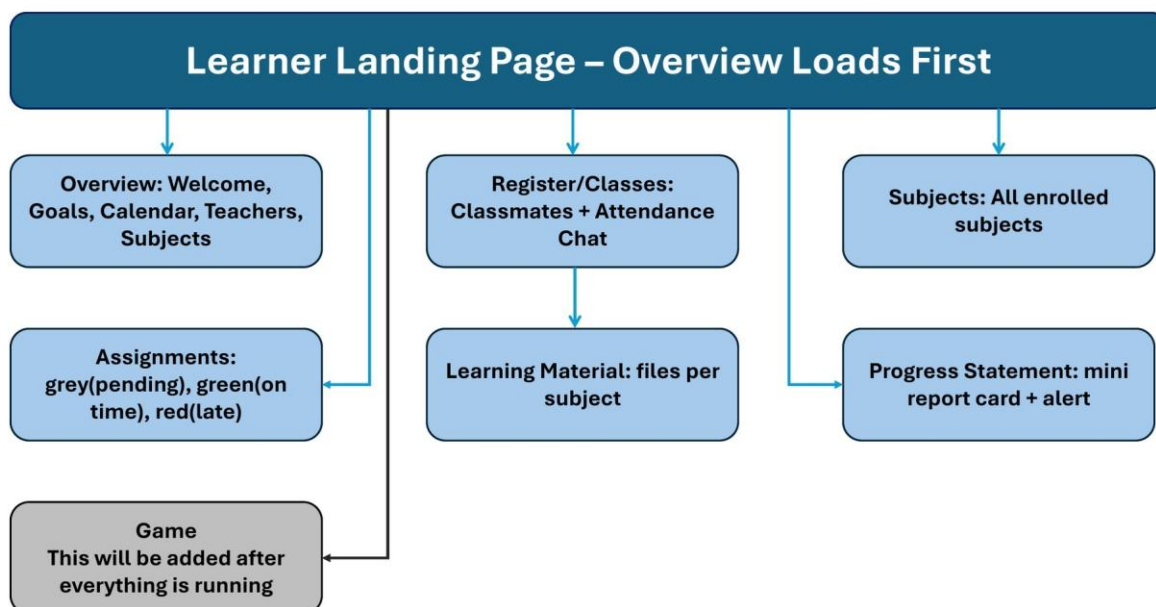
The diagrams below trace how a user moves through the application, from login through to each role's landing page and its main sections.

2.1 Login Flow



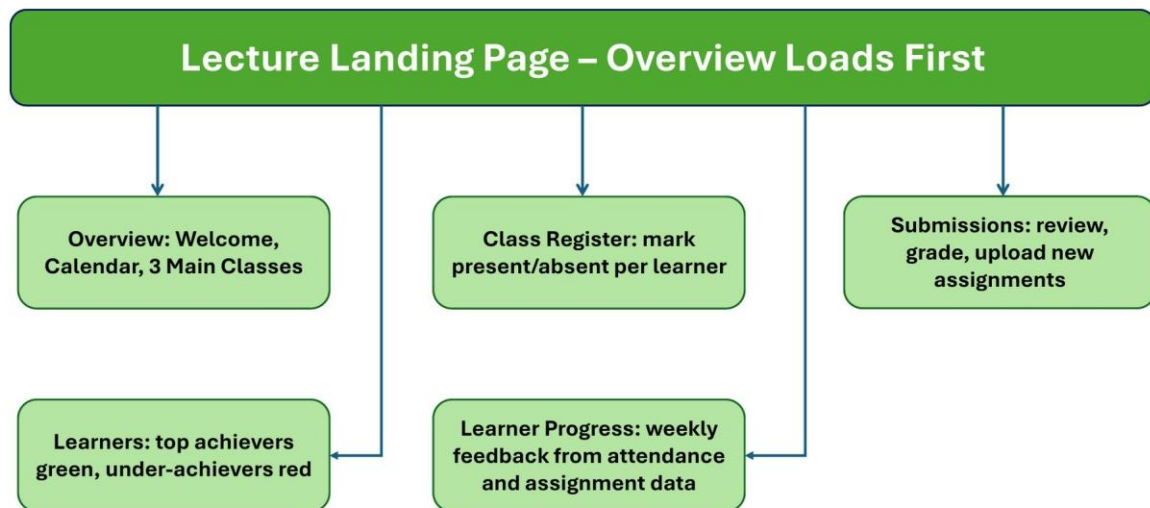
Both learners and assessors/lecturers share the same login screen. Three failed attempts or a forgotten password route the user through the reset flow before returning to Login.

2.2 Learner Landing Page



Once a learner signs in, the Overview loads first. From there they can reach Register/Classes, Subjects, Assignments, Learning Material and Progress Statement; Game is added once the core platform is stable.

2.3 Lecturer Landing Page



The assessor/lecturer sees a parallel Overview, built around Class Register, Submissions, Learners and Learner Progress feedback.

3. User Stories — Learner

Learner Login

User story: As a learner, I want to log in with my email and password so that I can access my own learning dashboard.

Acceptance criteria:

- Given three consecutive failed attempts, the learner is shown the 'Forgot password?' link in red.
- Given a registered learner account, entering a correct email and password signs the learner in and redirects to the Learner landing page.
- Given incorrect credentials, an inline error is shown, and the learner stays on the Login screen.

Password Reset

User story: As a learner, I want to reset my password so that I can regain access if I forget it.

Acceptance criteria:

- Clicking 'Forgot password?' sends a reset link/OTP to the registered email.
- Following the link lets the learner set a new password and return to Login.
- The old password stops working once the new one is confirmed.

Update Profile

User story: *As a learner, I want to update my login and personal details so that my profile reflects who I am.*

Acceptance criteria:

- The learner can edit name, photo, contact details and change their password from a Profile screen.
- Changes to email require re-authentication before they are saved.
- Saved changes are reflected across the app (header, register, submissions) without needing to log out.

Learner Landing Page (Overview)

User story: *As a learner, I want to land on my own Overview page after logging in so that I immediately see what matters to me today.*

Acceptance criteria:

- After successful login, the learner is taken to Overview, never to the Lecturer's landing page.
- The top bar shows the school logo (top-right), the learner's profile (top-left), and a notification bell next to it.
- The left sidebar lists, top to bottom: Overview, Register/Classes, Subjects, Assignments, Progress Statement, Learning Material, Game.

Overview — Daily Goals, Calendar, Teachers, Subjects

User story: *As a learner, I want an Overview with a welcome message, my own daily goals, a calendar, my teachers, and my main subjects so that I can plan my day as easily as possible.*

Acceptance criteria:

- A welcome statement greets the learner by name.
- A daily goals bar lets the learner add, edit, tick off and remove their own goals.
- A calendar widget shows the current month with class/assignment dates.
- The learner's assigned teachers are listed with subject and photo.
- Three of the learner's main subjects are highlighted as cards.

Register / Classes

User story: *As a learner, I want to see my classmates and my attendance trend so that I understand my class participation.*

Acceptance criteria:

- A list of classmates for the learner's classes is displayed.
- An attendance graph is rendered from register data captured by the assessor/lecturer on school days.
- The graph updates automatically when a new register is submitted.

Subjects

User story: *As a learner, I want to see every subject I am enrolled in so that I can navigate to their content quickly.*

Acceptance criteria:

- All enrolled subjects are listed with subject name, teacher, and a shortcut into Learning Material and Assignments for that subject.

Assignments with Status Colour-Coding

User story: *As a learner, I want assignments colour-coded by status so that I instantly know what is due, done, or overdue.*

Acceptance criteria:

- An assignment not yet due and not submitted is shown in grey, grouped by subject.
- An assignment submitted before or on the due date turns green.
- An assignment past its due date and not submitted turns red.
- Tapping an assignment opens its brief, due date and submission action.

Learning Materials

User story: *As a learner, I want learning materials organised by subject so that I can find what my teacher uploaded for that subject.*

Acceptance criteria:

- Subjects the learner is enrolled in are listed; opening one shows only material the lecturer uploaded for that subject.
- Material can be previewed or downloaded.

Progress Statement

User story: *As a learner, I want a running progress statement so that I can see how I am doing and whether I need extra help.*

Acceptance criteria:

- The statement is generated from class attendance and assignment completion/marks.
- If performance is below average, the statement clearly states the learner needs extra help and in which area.
- The learner can view a history of past progress statements.

Game

User story: *As a learner, I want a game in my sidebar so that I have an engaging, rewarding activity once the core platform is stable.*

Acceptance criteria:

- The Game item appears in the sidebar but is marked 'coming soon' / locked until it is built.
- It is the last item in the sidebar, after Learning Material.

4. User Stories — Assessor / Lecturer

Lecture/Assessor Login

User story: *As an assessor/lecturer, I want to log in using my email and password to access my teaching dashboard.*

Acceptance criteria:

- Correct credentials sign the assessor in and redirect to the Assessor/Lecturer landing page, never the learner layout.
- Incorrect credentials show an inline error and keep the assessor on the Login screen.

Password Reset

User story: *As an assessor/lecturer, I want to reset my password so that I can regain access if I forget it.*

Acceptance criteria:

- Clicking 'Forgot password?' sends a reset link/OTP to the registered email.
- Following the link lets the assessor set a new password and return to Login.
- The old password stops working once the new one is confirmed.

Update Profile

User story: *As an assessor/lecturer, I want to update my login and personal details so that my profile reflects who I am.*

Acceptance criteria:

- The assessor can edit name, photo, contact details and password from a Profile screen.
- Subjects/classes taught remain linked to the account after any profile edit.

Lecturer Landing Page (Overview)

User story: *As an assessor/lecturer, I want to land on my own Overview after logging in so I see my teaching day very easily.*

Acceptance criteria:

- The layout matches the learner shell (logo top-right, profile top-left, notifications) but the sidebar differs.
- The left sidebar lists, top to bottom: Overview, Class Register, Submissions, Learners, Learner Progress Feedback.

Overview — Calendar and Main Classes

User story: *As an assessor/lecturer, I want a welcome-back message, my calendar and my three main classes so I can plan my teaching day.*

Acceptance criteria:

- A 'Welcome back' statement greets the assessor by name.
- A calendar shows upcoming classes, due dates and register days. • The three main classes taught are shown as quick-access cards.

Class Register / Classes

User story: *As an assessor/lecturer, I want to mark attendance per class so that learner attendance data is accurate and up to date.*

Acceptance criteria:

- All classes the assessor teaches are listed; opening one lists every enrolled learner.
- Each learner can be marked present or absent (and optionally late) for that session.
- Submitting the register writes attendance to the learner's profile and instantly feeds the learner's Register/Classes attendance chart.

Submissions

User story: *As an assessor/lecturer, I want to review submitted assignments so I can grade them and keep learners informed.*

Acceptance criteria:

- All submitted assignments are listed with the submitting learner's profile.
- The assessor can leave written feedback and a grade per submission.
- The assessor can upload a new assignment (brief, due date, subject, class, attachment) from this screen.

Learners

User story: *As an assessor/lecturer, I want to see all the learners I teach so I can monitor performance as easily as possible.*

Acceptance criteria:

- Every learner taught by the assessor is listed with a shortcut into their profile.
- Top achievers are highlighted in green; under-achievers are highlighted in red, based on progress data.
- The assessor can leave feedback directly on a learner's profile from this screen.

Learner Progress Feedback

User story: *As an assessor/lecturer, I want to give weekly progress feedback, so learners and guardians know where they stand.*

Acceptance criteria:

- Feedback is composed using class attendance, assignment progress, and the learners list as source data.
- Feedback is saved to the learner's progress statement and timestamped.
- Learners who fall below average are flagged so the assessor is prompted to prioritise feedback for them.

5. REST API Design

Endpoints sit as a thin Cloud Functions / REST layer in front of Firebase Auth and Firestore. Paths in the Endpoint column refer to the Firestore document paths in the data model (Section 6). All endpoints require a valid Firebase ID token, except `/auth/login` and `/auth/forgot-password`.

5.1 Authentication & Profile

Method	Endpoint	Description	Role Access
POST	<code>/auth/login</code>	Email + password sign-in for learner or assessor; returns ID token and role.	Public
POST	<code>/auth/forgot-password</code>	Sends a reset link/OTP to the registered email.	Public
POST	<code>/auth/reset-password</code>	Confirms OTP/token and sets a new password.	Public

GET	/users/{uid}	Fetch a profile (own profile, or a taught learner's for an assessor).	Learner/Lecture
PATCH	/users/{uid}	Update personal details (name, phone, photo).	Learner/Lecture

5.2 Learner Overview — Goals, Teachers, Classmates, Attendance

Method	Endpoint	Description	Role Access
GET	/users/{uid}/goals	List the learner's daily goals.	Learner
POST	/users/{uid}/goals	Create a new daily goal.	Learner
PATCH	/users/{uid}/goals/{goalId}	Edit or complete a goal.	Learner
DELETE	/users/{uid}/goals/{goalId}	Remove a goal.	Learner
GET	/users/{uid}/teachers	List the learner's assigned teachers.	Learner
GET	/users/{uid}/classmates	List classmates across the learner's classes.	Learner
GET	/bookings?classId=&learnerId=	Attendance records used to build the attendance chart.	Learner, Lecture

5.3 Assignments & Submissions

Method	Endpoint	Description	Role Access
GET	/tasks?subjectId=&classId=	List assignments for a subject/class.	Learner, Lecture
GET	/tasks/{taskId}	Fetch a single assignment's detail.	Learner, Lecture
POST	/tasks	Create a new assignment (brief, due date, attachment).	Lecture
PATCH	/tasks/{taskId}	Edit an assignment.	Lecture
POST	/tasks/{taskId}/submissions	Learner submits work for an assignment.	Learner
Method	Endpoint	Description	Role Access
GET	/tasks/{taskId}/submissions	List submissions for grading.	Lecture

5.4 Class Register / Attendance

Method	Endpoint	Description	Role Access
GET	/bookings?assessorId=&date=	List an assessor's classes for a given day.	Lecture
POST	/bookings	Create a register/booking entry for a class session.	Lecture
POST	/bookings/{bookingId}/attendance	Mark present/absent/late per learner.	Lecture

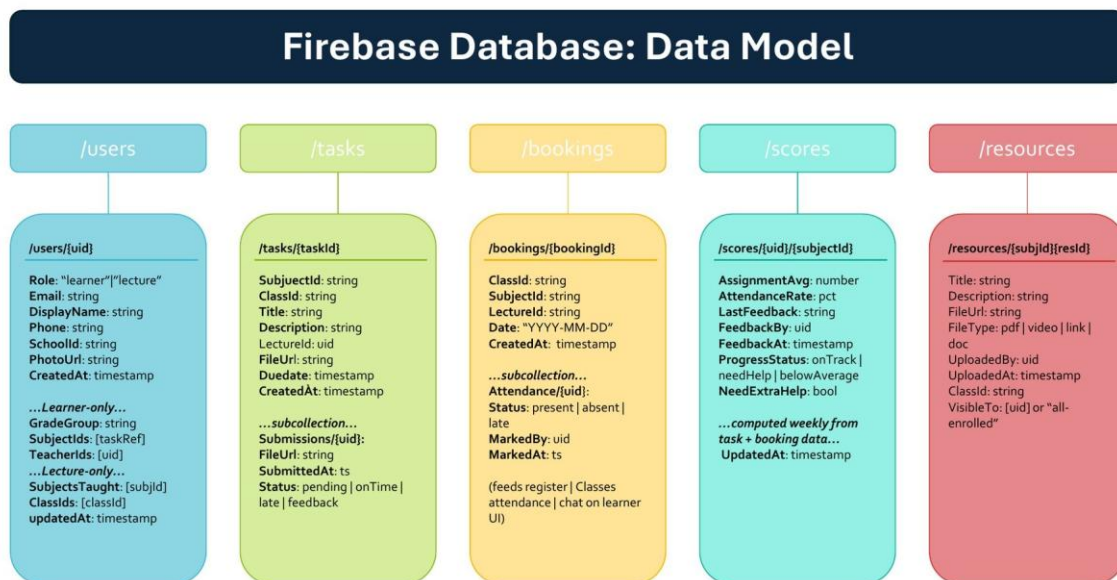
GET	/bookings/{bookingId}/attendance	Read the attendance list for a session.	Lecture
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5.5 Learners, Progress & Resources

Method	Endpoint	Description	Role Access
GET	/assessors/{uid}/learners	List all learners an assessor teaches, with rank flags.	Lecture
GET	/scores/{uid}	Fetch a learner's progress statement across subjects.	Learner, Lecture
POST	/scores/{uid}/{subjectId}/feedback	Post weekly progress feedback for a learner.	Lecture
GET	/resources?subjectId=	List learning material for a subject.	Learner, Lecture
POST	/resources	Upload new learning material for a subject.	Lecture
GET	/notifications?uid=	List notifications for the notification bell.	Learner, Lecture

6. Firebase Database — Data Model

Firestore is organised into five top-level collections: `/users`, `/tasks`, `/bookings`, `/scores` and `/resources`. Attendance and submissions live as subcollections beneath `/bookings` and `/tasks` so each record can be traced back to its parent session or assignment.

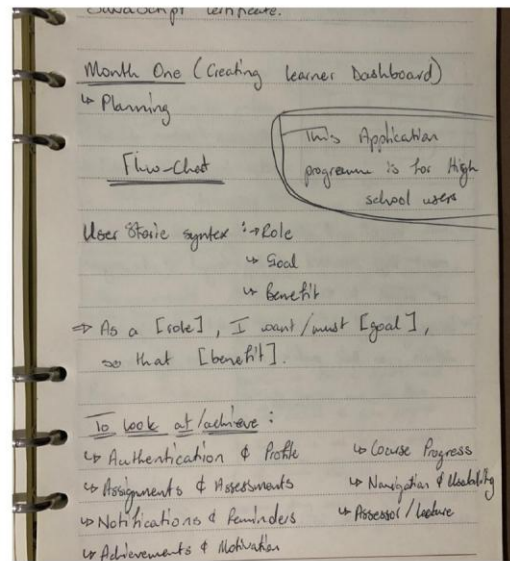
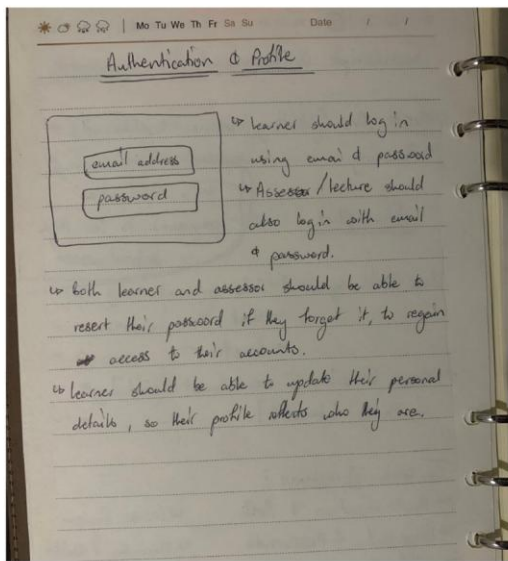


Firestore data model — top-level collections and their key fields.

7. Rough Sketches — Early Planning Notes

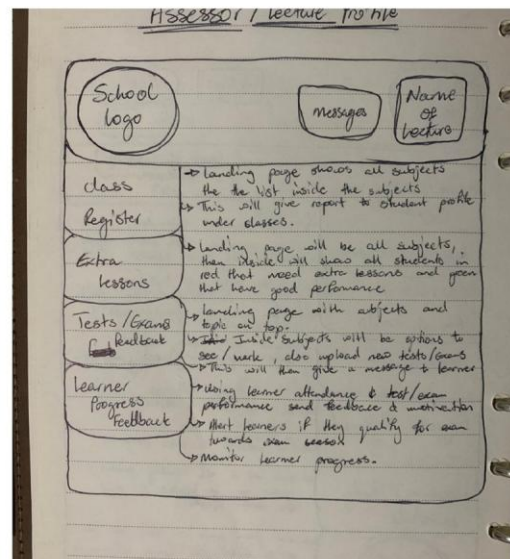
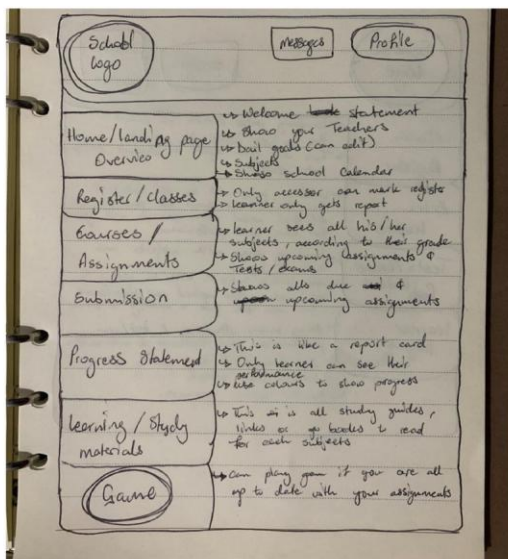
These are the original hand-drawn notes and wireframes that the user flow diagrams, user stories and API design above were developed from.

7.1 Planning Notes



Left: authentication and profile requirements. Right: Month 1 planning scope and the user-story syntax used throughout this document.

7.2 Wireframe Sketches



Left: Learner dashboard layout — home/overview, register/classes, courses/assignments, submissions, progress statement, learning materials, game. Right: Assessor/Lecturer dashboard layout — class register, extra lessons, tests/grades feedback, learner progress feedback.